

ALA :1

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Course: B.Tech Information Technology

Subject: Cloud Computing (BETCE17336)

ALA 1

Cloud Service Deployment: In this activity, each student will create and deploy a basic web application using a cloud platform to understand real-world cloud computing concepts. They will configure compute services (such as virtual machines or app services) and storage services (such as cloud storage or databases), and demonstrate successful application hosting. The implementation should showcase deployment, accessibility over the internet, and basic resource management. Students will submit the source code, deployment details, and a screenshot of the running application, and upload them on the GMIU portal

Solution

1. Objective

To deploy a basic web application using cloud services and understand practical concepts of cloud computing, including application deployment, compute services, storage services, internet accessibility, and basic resource management.

2. Project Overview

Vishva Parmar's Cloud Computing ALA-1 website is a web project published using GitHub Pages. It demonstrates how a static website can be hosted and accessed through the internet.

The website is built using web technologies and is publicly accessible through its GitHub Pages URL.

Technologies Used

- HTML5
- CSS3
- JavaScript
- Git and GitHub
- GitHub Pages
- Hosting Platform: GitHub Pages.

- Storage: The website source files and related assets are maintained in the GitHub repository. Amazon S3 and Azure Blob Storage are not listed because they are not part of the stated deployment.

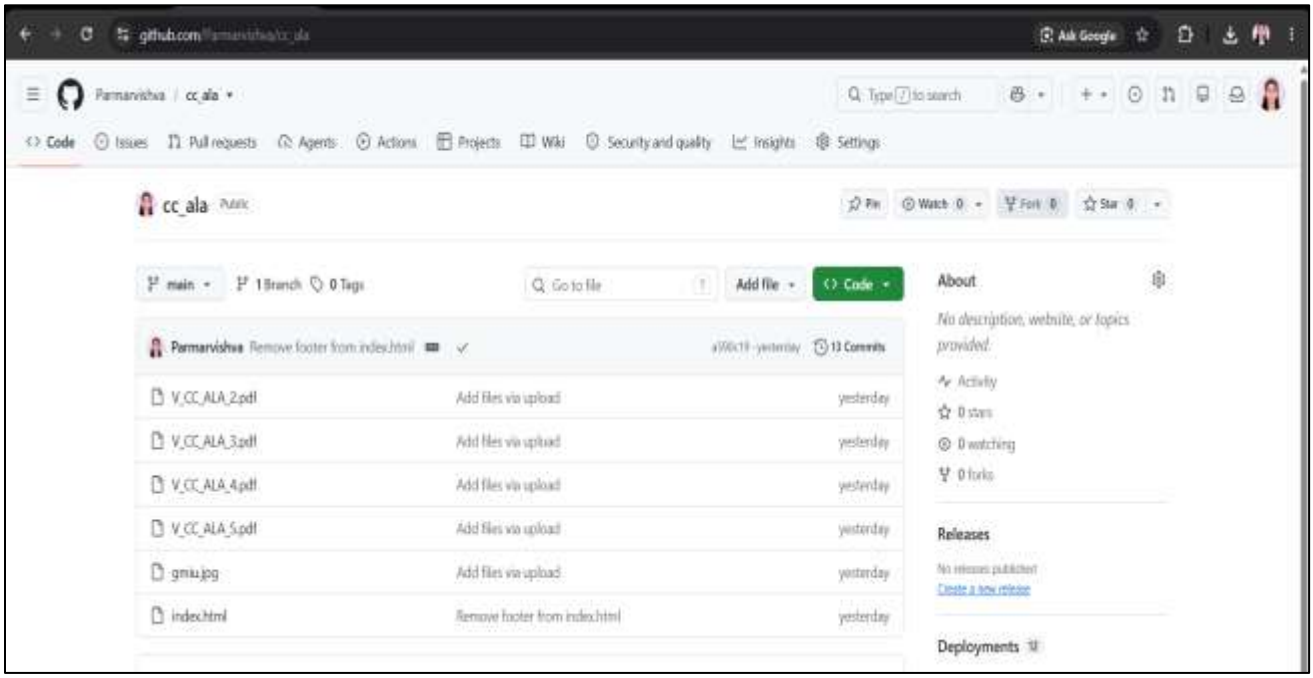
3. Source Code

The source code for the website is maintained in a GitHub repository. It contains the website files and any related resources used by the project.

Repository URL: https://github.com/parmarvishva/cc_ala

The source code repository provides version control and allows changes to the website to be maintained and deployed.

Figure 1: GitHub repository for the project



4. Deployment

The website was published using GitHub Pages and made available through a public URL.

The basic deployment process was:

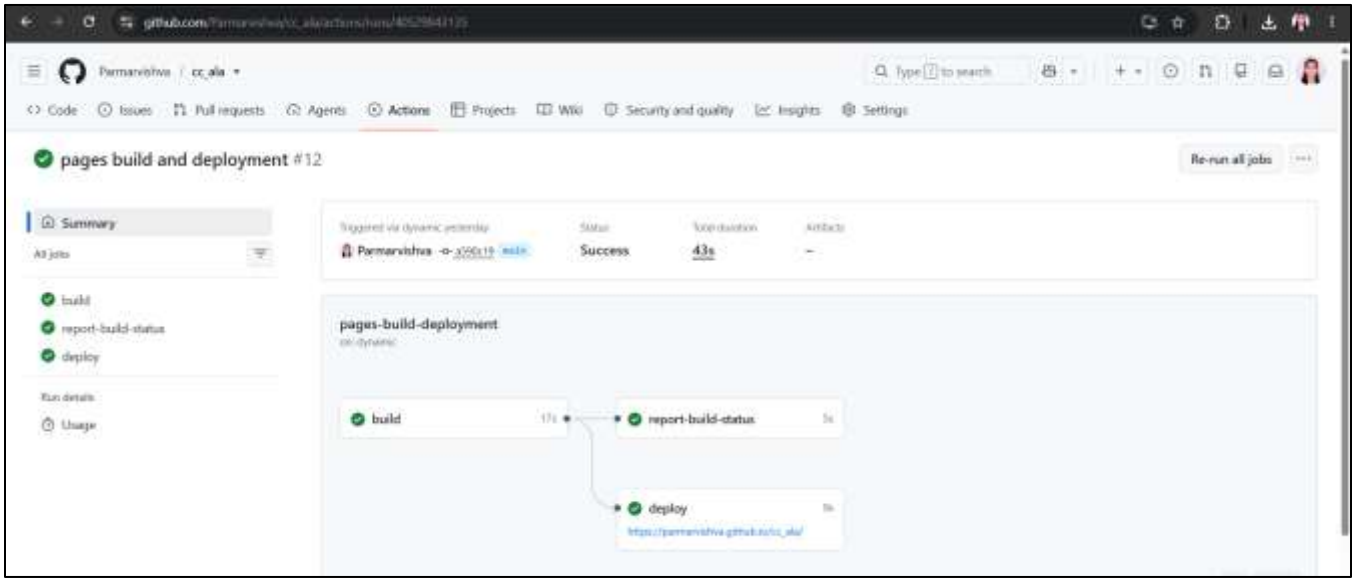
1. Website source code was maintained in GitHub.
2. GitHub Pages was configured as the website hosting platform.
3. The website files were published through the configured GitHub Pages deployment.
4. The deployment was tested using a web browser.

5. The application was made available through a public URL.

Hosting Platform: GitHub Pages.

Deployment Platform: GitHub pages

Figure 2: GitHub Pages deployment configuration



5. Compute Service

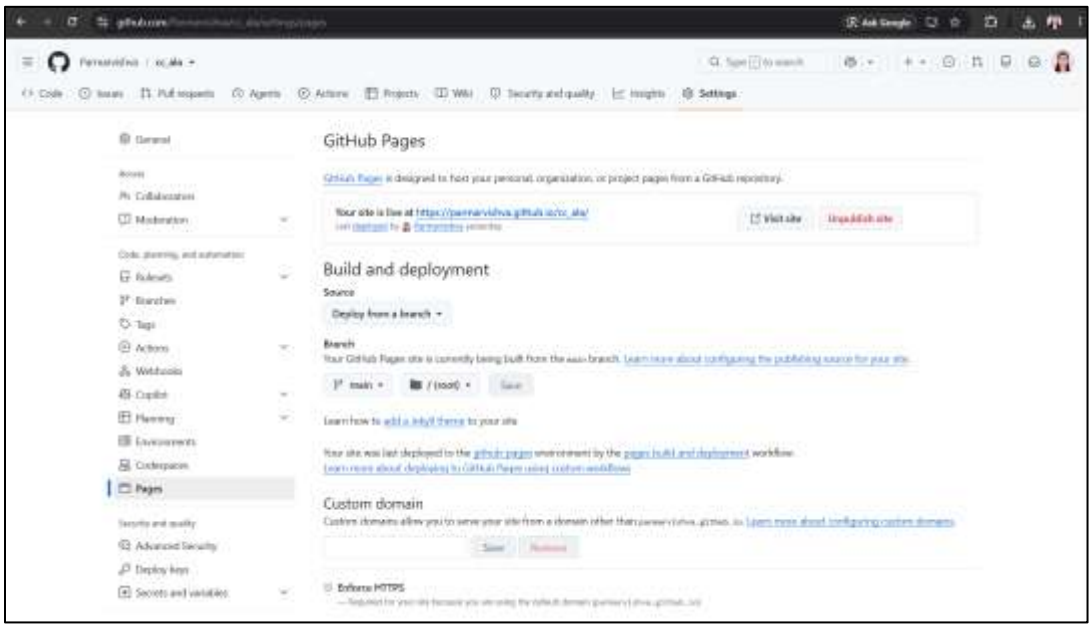
The compute service provides the environment required to host and serve the web application.

Hosting Service Used: GitHub Pages for static website hosting.

GitHub Pages serves the published static website when users open its public URL. No separate virtual machine or cloud compute instance is configured as part of this project.

This hosting service makes the website files available to visitors through a web browser.

Figure 3: GitHub Pages hosting service



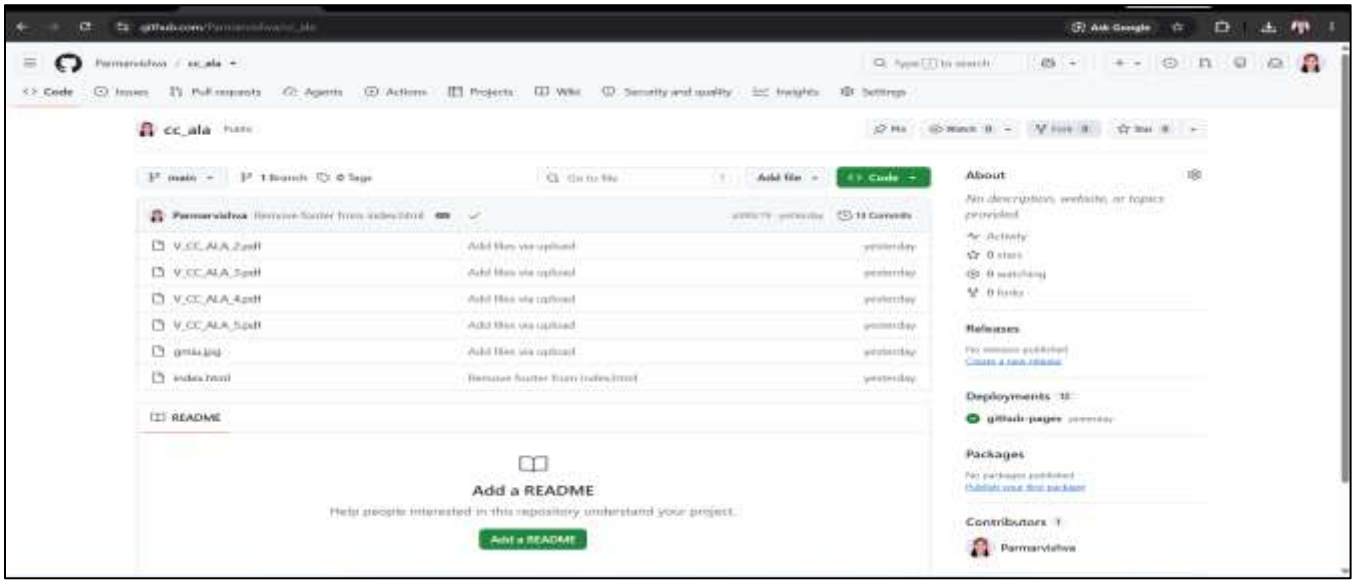
6. Storage Service

The website source files and required assets are stored in the project's GitHub repository.

Storage: The website source files and related assets are maintained in the GitHub repository. Amazon S3 and Azure Blob Storage are not listed because they are not part of the stated deployment.

The repository can contain HTML, CSS, JavaScript, images, and other files required by the website.

Figure 4: Project files stored in the GitHub repository



7. Internet Accessibility

The deployed website is publicly accessible through the internet.

Live Website URL:

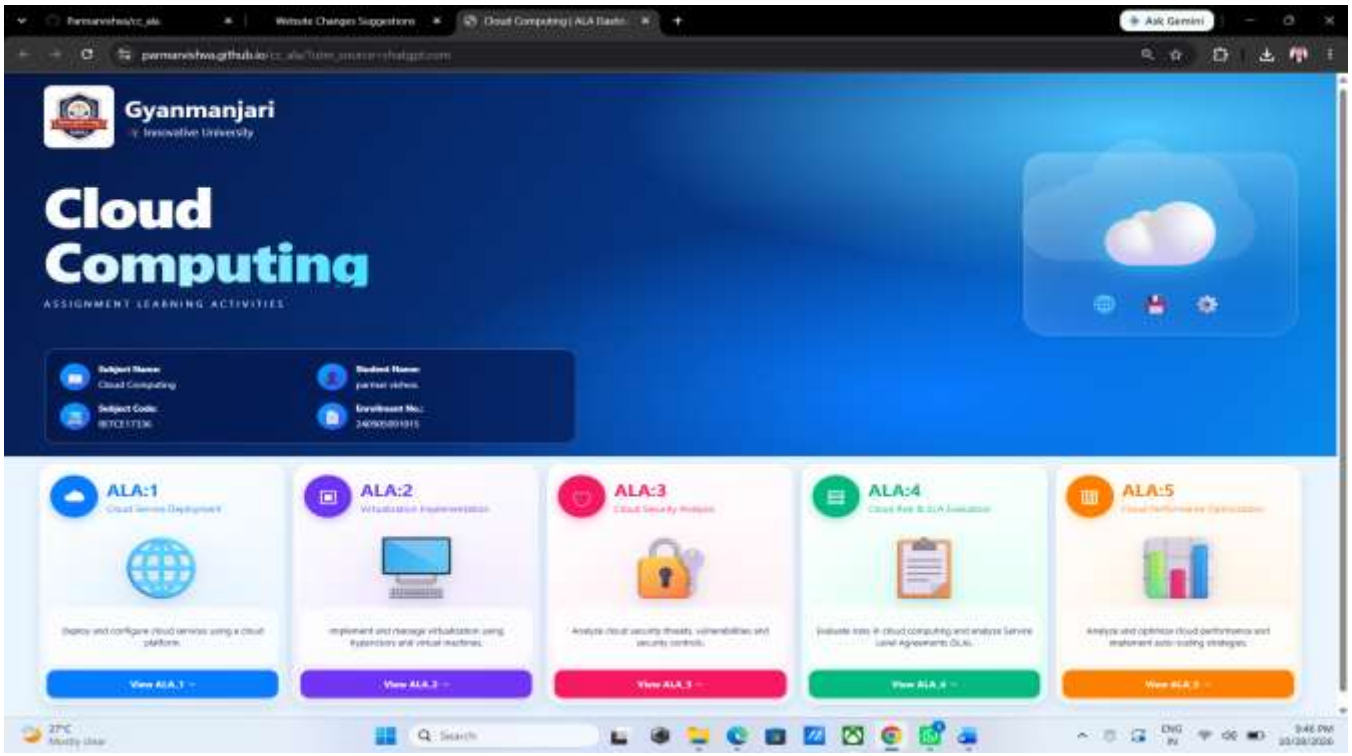
https://parmarvishva.github.io/cc_ala/

Users can access the website using a standard web browser without running the files locally.

The website presents the content and sections included in the Cloud Computing ALA-1 project.

- Home
- Project Overview
- Deployment Details
- Source Code
- Deployment Architecture
- Result
- Conclusion

Figure 5: Project website running successfully over the internet



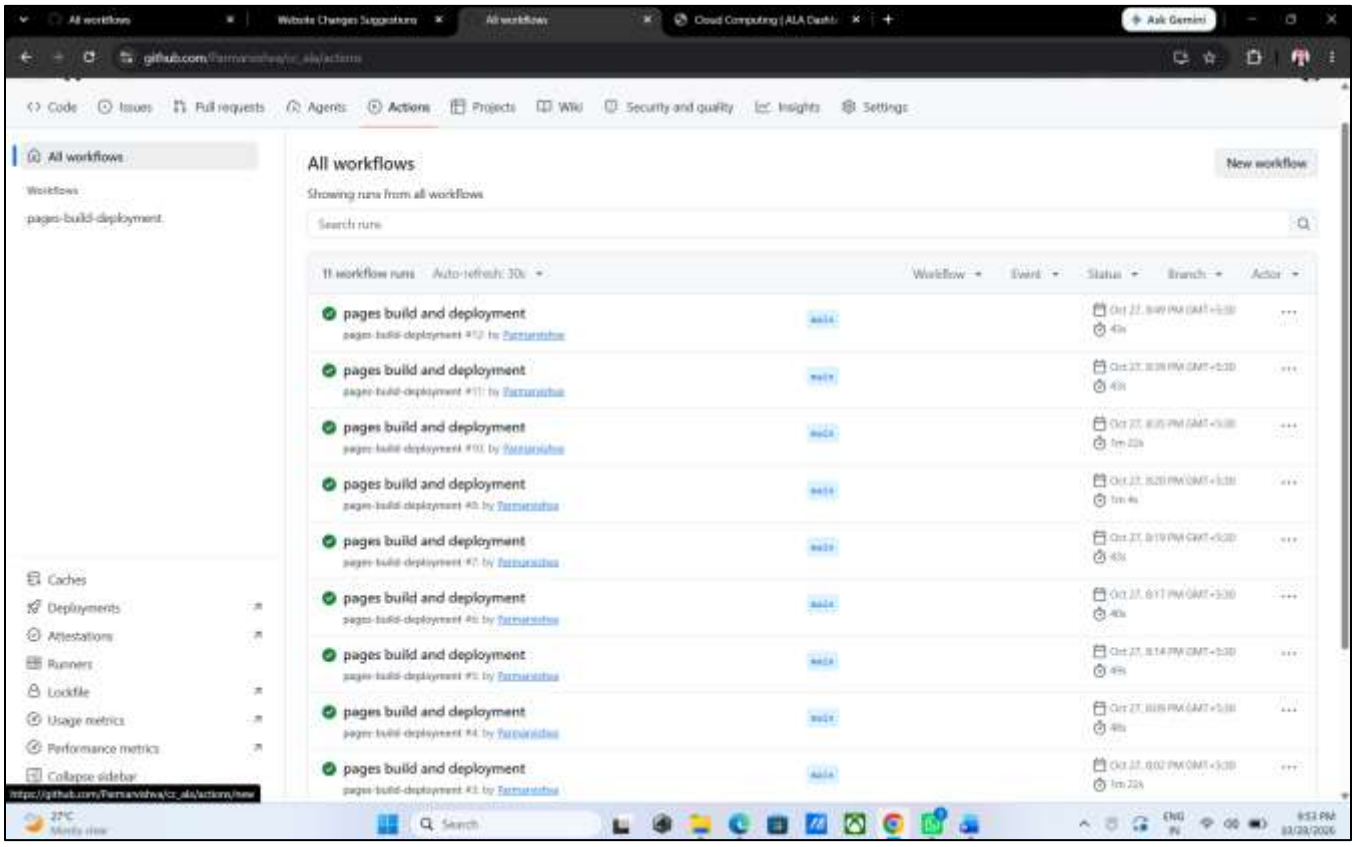
8. Resource Management

Basic resource management is performed through the GitHub repository and GitHub Pages settings. The deployment status and website files can be checked and updated when required.

The management activities include:

- Checking the GitHub Pages deployment status
- Managing website source files in the repository
- Updating website files and assets
- Checking that the public website URL is accessible
- Committing changes and redeploying the website when required

Figure 6: GitHub repository and deployment management



9. Deployment Architecture

Deployment architecture diagram to be added here: User → Web Browser → GitHub Pages → Website Files



10. Result

The website is published through GitHub Pages and can be accessed through its public URL. This project demonstrates source-code management, static website hosting, deployment, internet accessibility, and basic website maintenance.

The deployed website can be opened and tested using a standard web browser.

11. Conclusion

This activity provided practical experience in publishing a static website using GitHub Pages. The project demonstrates source-code management, deployment, internet accessibility, and basic resource management.

The activity helped in understanding how a website can be published and made available to users through a public hosting platform.